

1. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle A has mass 3 kg and a particle B has mass 2 kg .

The particles are moving on a smooth horizontal plane when they collide directly.

Immediately **before** the collision, the velocity of A is $(3\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$ and the velocity of B is $(-6\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$

Immediately **after** the collision the velocity of A is $\left(-2\mathbf{i} + \frac{2}{3}\mathbf{j}\right)\text{ m s}^{-1}$

- (a) Find the total kinetic energy of the two particles **before** the collision. (3)
- (b) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the impulse exerted on A by B in the collision. (3)
- (c) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of B immediately **after** the collision. (3)

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1.

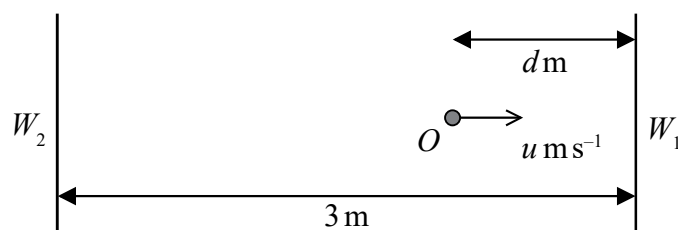


Figure 1

Figure 1 represents the plan of part of a smooth horizontal floor, where W_1 and W_2 are two fixed parallel vertical walls. The walls are 3 metres apart.

A particle lies at rest at a point O on the floor between the two walls, where the point O is d metres, $0 < d \leq 3$, from W_1 .

At time $t = 0$, the particle is projected from O towards W_1 with speed $u\text{ ms}^{-1}$ in a direction perpendicular to the walls.

The coefficient of restitution between the particle and each wall is $\frac{2}{3}$.

The particle returns to O at time $t = T$ seconds, having bounced off each wall once.

(a) Show that $T = \frac{45 - 5d}{4u}$ (6)

The value of u is fixed, the particle still hits each wall once but the value of d can now vary.

(b) Find the least possible value of T , giving your answer in terms of u . You must give a reason for your answer. (2)



1. A particle A of mass $3m$ and a particle B of mass m are moving along the same straight line on a smooth horizontal surface. The particles are moving in opposite directions towards each other when they collide directly.

Immediately before the collision, the speed of A is ku and the speed of B is u .

Immediately after the collision, the speed of A is v and the speed of B is $2v$.

The magnitude of the impulse received by B in the collision is $\frac{3}{2}mu$.

- (a) Find v in terms of u only.

(3)

- (b) Find the two possible values of k .

(5)

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3. A particle of mass m kg lies on a smooth horizontal surface.

Initially the particle is at rest at a point O between two fixed parallel vertical walls.

The point O is equidistant from the two walls and the walls are 4 m apart.

At time $t = 0$ the particle is projected from O with speed u m s⁻¹ in a direction perpendicular to the walls.

The coefficient of restitution between the particle and each wall is $\frac{3}{4}$

The magnitude of the impulse on the particle due to the first impact with a wall is λmu N s.

- (a) Find the value of λ .

(3)

The particle returns to O , having bounced off each wall once, at time $t = 7$ seconds.

- (b) Find the value of u .

(5)

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3. A particle P of mass $2m$ is moving in a straight line with speed $3u$ on a smooth horizontal plane. It collides directly with a particle Q of mass m that is moving on the plane with speed $2u$ in the opposite direction to P .

The coefficient of restitution between P and Q is e , where $e > \frac{4}{5}$

- (a) Show that the speed of Q immediately after the collision is $\frac{(4 + 10e)u}{3}$ (6)

After the collision Q hits a smooth fixed vertical wall that is perpendicular to the direction of motion of Q . The coefficient of restitution between Q and the wall is f .

- (b) Find, **in terms of e** , the set of values of f for which there will be a second collision between P and Q . (4)

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2. Two particles, A and B , are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.

Particle A has mass $5m$ and particle B has mass $3m$.

The coefficient of restitution between A and B is e , where $e > 0$

Immediately **after** the collision the speed of A is v and the speed of B is $2v$.

Given that A and B are moving in the same direction after the collision,

- (a) find the set of possible values of e .

(8)

Given also that the kinetic energy of A immediately after the collision is 16% of the kinetic energy of A immediately before the collision,

- (b) find

- (i) the value of e ,
 (ii) the magnitude of the impulse received by A in the collision, giving your answer in terms of m and v .

(6)

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5. Two particles, P and Q , are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.

The mass of P is $3m$ and the mass of Q is $4m$.

Immediately before the collision the speed of P is $2u$ and the speed of Q is u .

The coefficient of restitution between P and Q is e .

- (a) Show that the speed of Q immediately after the collision is $\frac{u}{7}(9e + 2)$

(6)

After the collision with P , particle Q collides directly with a fixed vertical wall and rebounds. The wall is perpendicular to the direction of motion of Q .

The coefficient of restitution between Q and the wall is $\frac{1}{2}$

- (b) Find the complete range of possible values of e for which there is a second collision between P and Q .

(4)

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5. A particle P of mass $3m$ and a particle Q of mass $2m$ are moving along the same straight line on a smooth horizontal plane. The particles are moving in opposite directions towards each other and collide directly.

Immediately before the collision the speed of P is u and the speed of Q is $2u$.

Immediately after the collision P and Q are moving in opposite directions.

The coefficient of restitution between P and Q is e .

- (a) Find the range of possible values of e , justifying your answer.

(8)

Given that Q loses 75% of its kinetic energy as a result of the collision,

- (b) find the value of e .

(3)

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3. Two particles, A and B , have masses $3m$ and $4m$ respectively. The particles are moving in the same direction along the same straight line on a smooth horizontal surface when they collide directly. Immediately before the collision the speed of A is $2u$ and the speed of B is u .

The coefficient of restitution between A and B is e .

- (a) Show that the direction of motion of each of the particles is unchanged by the collision.

(8)

After the collision with A , particle B collides directly with a third particle, C , of mass $2m$, which is at rest on the surface.

The coefficient of restitution between B and C is also e .

- (b) Show that there will be a second collision between A and B .

(6)

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4. A particle A of mass $2m$ is moving in a straight line with speed $3u$ on a smooth horizontal plane. Particle A collides directly with a particle B of mass m which is at rest on the plane.

The coefficient of restitution between A and B is e , where $e > 0$

- (a) Show that the speed of B immediately after the collision is $2u(1 + e)$. (6)

After the collision, B hits a smooth fixed vertical wall which is perpendicular to the direction of motion of B .

- (b) Show that there will be a second collision between A and B . (3)

The coefficient of restitution between B and the wall is $\frac{1}{2}$

Find, **in simplified form**, in terms of m , u and e ,

- (c) the magnitude of the impulse received by B in its collision with the wall, (3)

- (d) the loss in kinetic energy of B due to its collision with the wall. (3)

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8. A particle P of mass $2m$ and a particle Q of mass $5m$ are moving along the same straight line on a smooth horizontal plane.

They are moving in opposite directions towards each other and collide directly.

Immediately before the collision the speed of P is $2u$ and the speed of Q is u .

The direction of motion of Q is reversed by the collision.

The coefficient of restitution between P and Q is e .

- (a) Find the range of possible values of e .

(8)

Given that $e = \frac{1}{3}$

- (b) show that the kinetic energy lost in the collision is $\frac{40mu^2}{7}$.

(5)

- (c) Without doing any further calculation, state how the amount of kinetic energy lost in the collision would change if $e > \frac{1}{3}$

(1)

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